

Multicalibration for Unbiased Model-Based Prevalence Estimation

Fridolin Linder¹ Thomas Leeper¹ Daniel Haimovich¹ Niek Tax¹
Lorenzo Perini¹ Milan Vojnovic^{1,2}

¹Meta Platforms Inc., ²The London School of Economics and Political Science

Abstract

Political scientists increasingly use classifiers, and now large language models (LLMs), to estimate how prevalent a concept is in a corpus, such as topics in legislative agendas or categories in open-ended survey responses. Standard validation reports discriminative metrics such as accuracy or the area under the ROC curve (AUC) on a labeled sample. This certifies neither valid prevalence estimates in the validated population nor transportability to new ones. Discriminative metrics are blind to the feature-conditional errors that bias a prevalence estimate, and existing corrections either assume label shift or must be re-estimated for every target. Building on Kim et al. (2022), we show that a *multicalibrated* classifier, one calibrated conditional on input features rather than only on average, yields unbiased prevalence estimates under any covariate shift along the calibrated features within their support, and needs to be fitted only once. We explain why standard quantification methods fail where multicalibration does not, and evaluate it in a simulation and two applications: an LLM coding policy topics across four countries and languages, and a logistic regression predicting employment from survey data. Multicalibration keeps bias below half a percentage point for shifts within the calibrated features, where standard corrections miss by up to 19 points, and degrades only modestly beyond them.

Introduction

Machine learning classifiers, and increasingly large language models (LLMs), have become standard tools to estimate the prevalence of a concept in political text corpora (Baumgartner et al. 2006; Benoit et al. 2026; Overos et al. 2024; King and Lowe 2003; Weidmann et al. 2026; Mellon et al. 2024; Gilardi et al. 2023; Ornstein et al. 2025). The quantity of interest is usually not the label on any single document but an *aggregate* count or prevalence (González et al. 2017; Hopkins and King 2010).

The standard way to validate such a classifier is to measure how well it discriminates: how closely its labels agree with human codes, reported as accuracy, F_1 , or the area under the ROC curve (AUC) and judged against the intercoder reliability of the human benchmark (Halterman and Keith 2026; Hayes and Krippendorff 2007). For prevalence estimation this is not sufficient, for two related reasons. First, discrimination does not imply an unbiased prevalence estimate even on the population the classifier was validated on. A classifier can rank documents almost perfectly yet misstate the proportion that are positive, because its false positives and false negatives need

not cancel. The quantification literature corrects this, adjusting the raw count with error rates estimated on the labeled sample (González et al. 2017).

Second, that correction is tied to the population it was estimated on. When the same instrument is applied to a new target, for example to compare sub-corpora or track a quantity over time, it no longer holds in general. This is due to covariate shift. Across targets, the mix of documents changes (languages, venues, eras, authors) while the rule linking content to category stays fixed (e.g., through the codebook). When a target holds more of the cases the classifier misjudges, errors that canceled on the validation sample no longer do. Classical quantifiers rely on the opposite kind of stability: that documents of a category look alike everywhere and only their overall frequency changes (label shift), or at least that the classifier’s error rates within each category do not change. Neither is likely across languages and venues. A high AUC does not protect against this, and standard validation will not reveal it: validity established in one context need not hold in another (Adcock and Collier 2001).

Existing tools each address part of the problem, but none yields a *reusable* instrument. Quantification methods estimate prevalence from imperfect classifiers (González et al. 2017). Classify & Count takes the classifier’s labels at face value; its bias-corrected relatives (Rogan and Gladen 1978; Saerens et al. 2002) and the classifier-free ReadMe family (Hopkins and King 2010; Jerzak et al. 2023) instead carry class-conditional quantities, such as error rates or word profiles within each category, from the labeled sample to the target. Covariate shift generally changes those quantities (Tasche 2023). Design-based supervised learning (Egami et al. 2023) and prediction-powered inference (Angelopoulos et al. 2023) require gold labels drawn from the population being analyzed, and importance weighting (Shimodaira 2000; Sugiyama et al. 2007) must be re-estimated for each target. A related literature on learned proxies and text measures (Knox et al. 2022; Rodriguez and Spirling 2022) and on LLMs as annotators or substitutes for human data (Ziems et al. 2024; Argyle et al. 2023; Bisbee et al. 2024) raises the same worry that imperfect measures distort inference; Weidmann et al. (2026) find that LLM coders deviate from experts in systematic directions.

The simplest model-based prevalence estimator just averages the model’s outputs (scores or labels). Such an estimator is unbiased if the classifier is *calibrated*, that is, if its outputs can be interpreted as predicted probabilities that match empirical frequencies in the population at hand. Calibration, however, is a property of one population and need not survive a shift. Empirically, it degrades under covariate shift even when accuracy is preserved (Yang et al. 2023). Confidence elicitation (Tian et al. 2023; Kadavath et al. 2022; Wang et al. 2023) aims only at this global calibration. The closest precedent, S. Wu and Resnick (2024), casts prevalence estimation as calibrating classifier outputs on a labeled sample and extrapolating to the target. Their extrapolation step assumes that the calibration curve or the class-conditional score distribution is stable, and a globally calibrated score keeps neither under covariate shift when its errors vary with the features. Building on Kim et al. (2022), we argue that *multicalibration*, calibration conditional on the input features rather than on average (Hébert-Johnson et al. 2018; Detommaso et al. 2024), fills this gap. A multicalibrated classifier is stable under reweighting of those features. Calibrated once on a labeled source sample, it returns unbiased prevalence estimates on any target whose shift the features capture and that lies within their support (Kim et al. 2022; J. Wu et al. 2024).

Prevalence Bias Reduction through Multicalibration

Consider a binary outcome $Y \in \{0, 1\}$, features X , and a predictor $h(X) \in [0, 1]$. The estimand is the prevalence $\pi^* = P^*(Y = 1)$ in a target population, estimated by averaging predictions over unlabeled target observations. We assume *covariate shift*: $P(X)$ may differ between source and target, while $P(Y | X)$ remains stable. We also require overlap: any region of the feature space that can occur in the target must also occur in the source. Covariate shift contrasts with *label shift*, which instead holds $P(X | Y)$ fixed while the prevalence of Y changes.¹ Under concept drift, where $P(Y | X)$ changes, target prevalence cannot generally be recovered without target labels or additional identifying assumptions.

Global calibration is sufficient for unbiased prevalence estimation in a given population. If $\mathbb{E}[Y | h(X) = p] = p$ for every prediction value p , then $\mathbb{E}[h(X)] = \mathbb{E}[Y]$. But global calibration in the source population need not survive a change in the distribution of X . To see why, partition the population into feature-defined groups G with weights w_G , and let $\epsilon_G = \mathbb{E}[h(X) - Y | X \in G]$ be the mean signed error in group G . If the target changes only the relative sizes of these groups, its population prevalence error is

$$\mathbb{E}^*[h(X)] - \pi^* = \sum_G w_G^* \epsilon_G.$$

Global calibration in the source population requires only that the source-weighted errors cancel. When the group weights change, the same errors need not cancel in the target.

Discriminative performance does not prevent this problem. AUC and other rank-based metrics are invariant to strictly increasing transformations of the scores, even though such transformations can substantially change prevalence estimates. Accuracy and F_1 are not rank-invariant, but they measure unsigned classification performance rather than the signed errors relevant to an aggregate proportion. A classifier can therefore perform well overall while making systematic errors in a group that is rare in the source and common in the target.

Existing corrections rely on different stability conditions. Rogan-Gladen transports source estimates of sensitivity and specificity; SLD and related methods assume prior shift and valid source posteriors; and ReadMe relies on stability of class-conditional feature distributions (Rogan and Gladen 1978; Saerens et al. 2002; Hopkins and King 2010). Global recalibration assumes that the score–outcome relationship transports (S. Wu and Resnick 2024), which covariate shift in X does not guarantee once X is reduced to a score (Tasche 2022). Importance weighting instead estimates a target-specific density ratio and reweights the labeled source sample toward the target (Shimodaira 2000; Sugiyama et al. 2007). It shares the covariate-shift assumption made below, but it does not use the classifier at all: the estimate is a weighted average of hand codes, it must be re-estimated for every target, and it has nothing to reweight where the target contains feature values absent from the source.

Multicalibration removes these group-level errors rather than relying on their cancellation. Let $\mathcal{G}$ contain all groups that can be formed from the calibrated features, including their interactions and intersections. A predictor f is multicalibrated with respect to $\mathcal{G}$ if $\mathbb{E}[Y | f(X) = v, X \in G] = v$ for every $G \in \mathcal{G}$ and prediction value v (Hébert-Johnson et al. 2018). Averaging over prediction values shows that f has zero mean signed error within every group in $\mathcal{G}$. A target shift is captured by $\mathcal{G}$ if it can be expressed as a reweighting of these groups. Because each group error is zero, changing

¹The causal heuristic behind these invariance assumptions traces to @scholkopf2012causal; we rely on the invariance, not the causal direction.

their weights cannot create aggregate error. Under covariate shift and overlap,

$$\mathbb{E}^*[f(X)] = \mathbb{E}^*[Y] = \pi^*.$$

This is the “universal adaptability” guarantee of Kim et al. (2022), who show that such an estimate competes with inverse propensity weighting under any propensity model the group class can express; zero mean error within each group (multi-accuracy) already suffices (SI Section S1.8). The same source-fitted predictor can therefore be reused for every target shift captured by the calibrated features, without estimating a new correction for each target and without seeing the target at all. Where importance weighting corrects the *sample* toward one target, multicalibration corrects the *classifier* once: the estimate averages bounded predictions over every target document, so no small set of observations can dominate it, and the classifier’s document-level signal is retained rather than discarded.

The guarantee is exact only when the target-to-source density ratio lies in the span of the group indicators. Otherwise it is approximate, with error governed by how well the calibrated groups approximate that ratio; with finite calibration data, multicalibration itself also holds only approximately (SI Section S1.8). Whether covariate shift and this representation condition are plausible requires substantive judgment. Overlap can be diagnosed, though not generally certified, using unlabeled target data. We obtain the corrected predictor using MCGrad (Tax et al. 2026), a post-hoc gradient-boosting multicalibrator.

Simulation

We first illustrate the mechanism in a minimal setting. Each document has a binary feature $X \in \{0, 1\}$ (for example, its language) and a continuous content signal $U \sim N(0, 1)$ independent of X . The outcome follows $P(Y = 1 \mid X, U) = \sigma(a_X + 1.5U)$ with $a_0 = -2$ and $a_1 = 1.5$, so $P(Y = 1 \mid X = 0) \approx 0.19$ and $P(Y = 1 \mid X = 1) \approx 0.75$. This relationship is the same in every population; only $P(X)$ shifts. The classifier’s score is $\sigma(a_X + 1.5U + \delta_X)$ with $\delta_0 = 0.8$ and $\delta_1 = 0$: it is calibrated for documents with $X = 1$ and overstates the probability for documents with $X = 0$. Scores are continuous and overlap across the two groups, and the classifier discriminates well (AUC ≈ 0.89). Each estimator is fit on a labeled calibration sample of 10,000 documents drawn at $P(X = 0) = 0.5$ and then applied, with its parameters fixed, to unlabeled targets with $P(X = 0)$ between 0.01 and 0.99. Global recalibration is isotonic regression of Y on the score; multicalibration is MCGrad on the score and X . We repeat this 50 times.

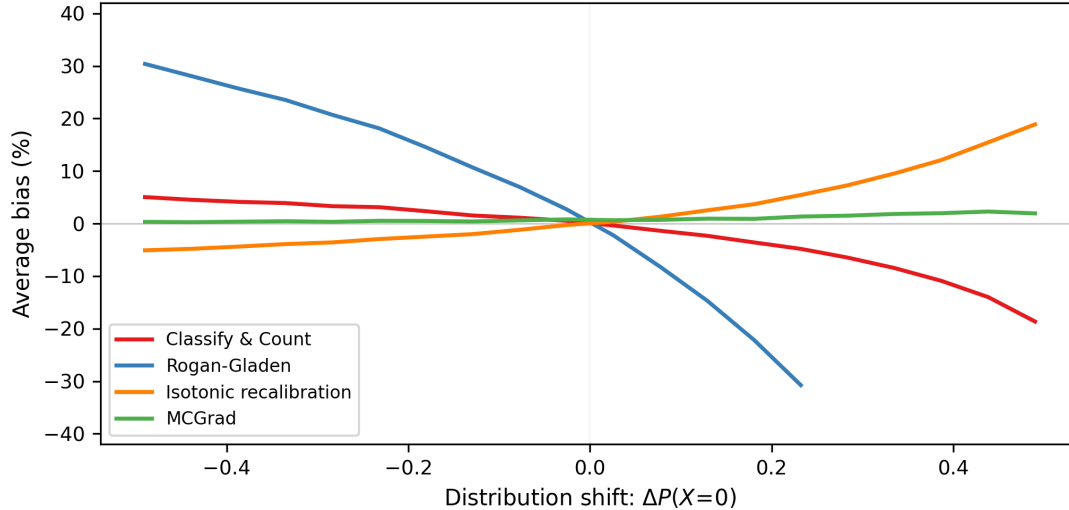


Figure 1. Relative prevalence bias under covariate shift (50 runs). The x-axis is the change in $P(X = 0)$ from the training value of 0.5. Classify & Count and Rogan-Gladen diverge with shift; isotonic recalibration is unbiased at the training distribution but drifts as the target shifts; MCGrad stays near zero. Cropped at $\pm 40\%$; the full range, additional methods, and RMSE are in SI Sections S4–S5.

At the training distribution ($\Delta P(X = 0) = 0$) every method is approximately unbiased (Figure 1). As the target shifts, they diverge. Classify & Count reaches about -19% at the most extreme shift, and Rogan-Gladen, whose ratio form amplifies the error, exceeds $\pm 30\%$. Isotonic recalibration is the instructive case. It makes the scores calibrated on the calibration sample, but it maps a given score to the same probability whatever the document’s X . That probability averages over the two groups at their calibration mix: too high for $X = 0$ documents and too low for $X = 1$ documents. The two errors cancel only at that mix, so the bias grows as the target moves away from it, to about $+19\%$ when nearly all documents have $X = 0$. MCGrad, which calibrates within each value of X , stays within about 2% throughout. All four estimators start from the same classifier scores; they differ only in how they turn those scores into a prevalence.

Applications

We demonstrate the mechanism on two applications with known ground truth. The first is an LLM coding political text. We use Claude Opus 4.6² to classify 30,000 Comparative Agendas Project texts (Baumgartner et al. 2006) for whether they concern Law & Crime (CAP major topic 12), scored against the published expert codes. The documents span six sub-populations across four countries and languages, 5,000 each. Law & Crime is a rare topic, coded in 7.9% of documents at baseline and 6–20% across scenarios. The second is an ordinary logistic regression predicting employment from 16 sociodemographic features in American Community Survey (ACS) microdata. In both, we construct targets whose covariate shift is known by design: within-support reweightings (of countries and document types; of age groups) and out-of-support targets (unseen document types; held-out states). The LLM is prompted for both a Yes/No classification and an elicited

²An open-weights replication with Llama 3.3 70B (SI Section S2) reproduces the qualitative pattern and permits full reproduction.

probability $P(\text{Yes})$ (Tian et al. 2023; Kadavath et al. 2022). We multicalibrate on the bare Yes/No label, the output essentially all applied studies produce, and reserve the probability for the methods that require a score. Designs, feature sets, and full numbers are in SI Sections S2–S3.

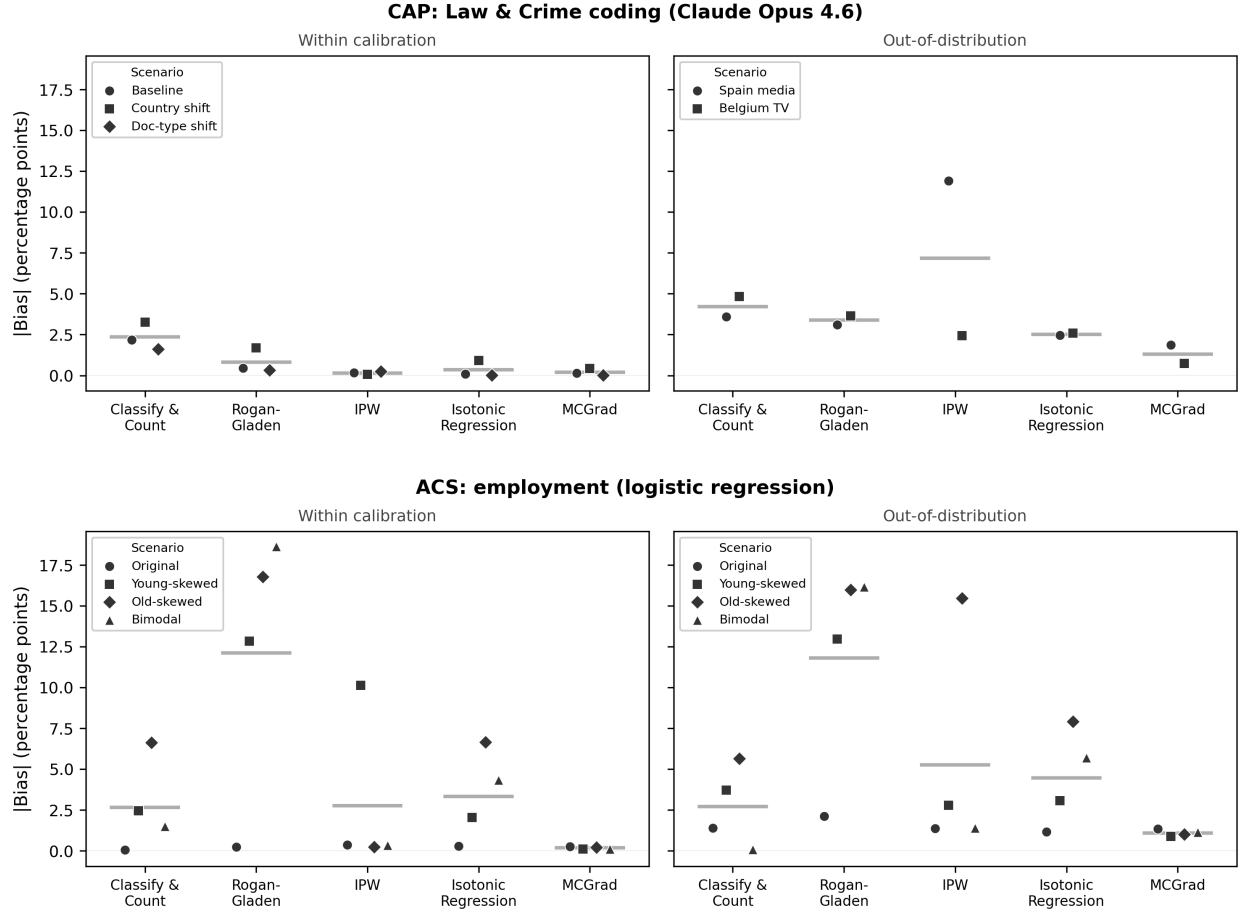


Figure 2. Absolute prevalence bias (percentage points) by method and scenario. Top: CAP Law & Crime coding with Claude Opus 4.6, multicalibrated on the Yes/No label. Bottom: ACS employment with logistic regression. Left: targets within the calibration support; right: out-of-distribution targets. Marker shape denotes scenario, horizontal bars are scenario means, and all panels share one scale. Full numbers in SI Tables S1–S2.

In the political-text application the LLM clears any discriminative-validation bar, with per-language AUC of 0.98–0.99, yet Classify & Count, the applied default, is biased by +1.6 to +4.8pp (Figure 2, top), a relative overstatement of nearly 30% at baseline and over 40% at its worst. Rogan-Gladen reduces this within support but still misses by +3.1 to +3.7pp out of support. SLD, the label-shift quantifier, run on isotonic-recalibrated scores so that its posteriors are valid in the source, is accurate at baseline (+0.1pp) but drifts under shift, to +1.4pp on the country shift and +3.8 to +4.7pp out of support, worse than isotonic recalibration alone: it attributes the covariate shift to a change in the class prior.³ Isotonic recalibration of the scores is accurate within support (≤ 0.9 pp) but drifts out of it (+2.6pp on Belgian TV, −2.5pp on Spanish media). Inverse probability weighting (IPW),

³ReadMe [@hopkinsking2010nonparametric], the canonical political-science quantifier, is classifier-free and fails as the label-shift diagnosis predicts, exceeding +60pp under some shifts; details and ReadMe2 [@jerkzackingstrezhnev2023improved] in SI Section S7 (Table S4).

with a gradient-boosted propensity model on the same features as multicalibration, is near-zero within support but fails out of it (-11.9pp on Spanish media), where the target contains feature values absent from the source and there are no hand codes to reweight. Multicalibration on the bare Yes/No labels achieves near-zero bias within support ($\leq 0.4\text{pp}$) and degrades gracefully out of support ($+0.7\text{pp}$ on Belgian TV, -1.9pp on Spanish media). Feeding it the elicited probabilities instead performs comparably within support and somewhat worse out of it (SI Table S2), because the correction comes from the metadata features and the calibration sample, not from the quality of the model’s confidences.

The survey application reproduces this ordering with a wider spread (Figure 2, bottom). Rogan-Gladen and SLD fail by $12\text{--}19\text{pp}$ under the age shifts. Classify & Count and isotonic regression stay closer but grow with the shift, reaching $6\text{--}8\text{pp}$ at the largest ones. IPW, with the same off-the-shelf learner and no tuning, is accurate on most targets ($\leq 0.4\text{pp}$ within support) but collapses on two (-10.1pp on the young-skewed shift and -15.5pp on the old-skewed one in held-out states): with a calibration sample thirty times larger than the target, a handful of source observations receive extreme weights and dominate the estimate (SI Section S3). Multicalibration, using the same learner, stays at $\leq 0.27\text{pp}$ across every within-support target and at $0.88\text{--}1.35\text{pp}$ on held-out states. Read together, the two applications locate the condition under which global recalibration is sufficient: it works when the predictor’s errors are roughly homogeneous across the groups a shift reweights, which the CAP shifts approximately satisfy. The ACS shifts move age, which drives employment hard (76% employed at ages 25–54 against 17% at 65+), so group errors are large and heterogeneous, and only feature-conditional calibration removes them.

Discussion

Prevalence estimation across populations is a common goal in political science research. With LLMs becoming more popular, even more zero-shot, model-based measurement of prevalence is being unlocked. Discriminative validation shows only that a classifier separates classes. It is silent on whether the resulting prevalence estimates are unbiased, either in the population where discrimination was validated or in any other. In our LLM application a classifier with per-language AUC above 0.98 still produced default prevalence estimates off by roughly 30–40% of the true rate, even without shift and worse under it, and would have passed any discriminative check while misreporting prevalence. Multicalibration, calibration conditional on features rather than on average, removes this bias where discrimination was validated and keeps it removed as the target shifts. Unlike importance weighting, which rests on the same assumption, it is fitted once, before any target is seen, and yields a single instrument that can be applied to a time series or a set of sub-corpora without re-estimation or per-target diagnostics.

However, the guarantee has limits. It holds only along calibrated dimensions, so calibration data must span the anticipated variation. It also requires labeled calibration data, which in zero-shot settings reintroduces some hand-coding, though far less than per-document labeling and only once. Under mixtures the covariate-shift condition holds only approximately. Our applications have ground truth, so we can verify the estimates directly; a practitioner usually cannot, since the target’s labels are what is missing.

References

- Adcock, Robert, and David Collier. 2001. “Measurement Validity: A Shared Standard for Qualitative and Quantitative Research.” *American Political Science Review* 95 (3): 529–46. <https://doi.org/10.1017/S0003055401003100>.
- Angelopoulos, Anastasios N., Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnica. 2023. “Prediction-Powered Inference.” *Science* 382 (6671): 669–74. <https://doi.org/10.1126/science.adi6000>.
- Argyle, Lisa P., Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. 2023. “Out of One, Many: Using Language Models to Simulate Human Samples.” *Political Analysis* 31 (3): 337–51. <https://doi.org/10.1017/pan.2023.2>.
- Baumgartner, Frank R., Christoffer Green-Pedersen, and Bryan D. Jones. 2006. “Comparative Studies of Policy Agendas.” *Journal of European Public Policy* 13 (7): 959–74. <https://doi.org/10.1080/13501760600923805>.
- Benoit, Kenneth, Scott De Marchi, Conor Laver, Michael Laver, and Jinshuai Ma. 2026. “Using Large Language Models to Analyze Political Texts Through Natural Language Understanding.” *American Journal of Political Science*, ahead of print. <https://doi.org/10.1111/ajps.70050>.
- Bisbee, James, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. 2024. “Synthetic Replacements for Human Survey Data? The Perils of Large Language Models.” *Political Analysis* 32 (4): 401–16. <https://doi.org/10.1017/pan.2024.5>.
- Detommaso, Gianluca, Martin Bertran, Riccardo Fogliato, and Aaron Roth. 2024. “Multicalibration for Confidence Scoring in LLMs.” *Proceedings of the 41st International Conference on Machine Learning (ICML)*. <https://proceedings.mlr.press/v235/detommaso24a.html>.
- Egami, Naoki, Musashi Hinck, Brandon M. Stewart, and Hanying Wei. 2023. “Using Imperfect Surrogates for Downstream Inference: Design-Based Supervised Learning for Social Science Applications of Large Language Models.” *Advances in Neural Information Processing Systems (NeurIPS)* 36: 68589–601. <https://doi.org/10.52202/075280-3000>.
- Gilardi, Fabrizio, Meysam Alizadeh, and Maël Kubli. 2023. “ChatGPT Outperforms Crowd-Workers for Text-Annotation Tasks.” *Proceedings of the National Academy of Sciences* 120 (30): e2305016120. <https://doi.org/10.1073/pnas.2305016120>.
- González, Pablo, Alberto Castaño, Nitesh V. Chawla, and Juan José Del Coz. 2017. “A Review on Quantification Learning.” *ACM Computing Surveys* 50 (5): 1–40. <https://doi.org/10.1145/3117807>.
- Halterman, Andrew, and Katherine A. Keith. 2026. “Codebook LLMs: Evaluating LLMs as Measurement Tools for Political Science Concepts.” *Political Analysis* 34 (2): 188–204. <https://doi.org/10.1017/pan.2025.10017>.

- Hayes, Andrew F., and Klaus Krippendorff. 2007. “Answering the Call for a Standard Reliability Measure for Coding Data.” *Communication Methods and Measures* 1 (1): 77–89. <https://doi.org/10.1080/19312450709336664>.
- Hébert-Johnson, Ursula, Michael P. Kim, Omer Reingold, and Guy N. Rothblum. 2018. “Multicalibration: Calibration for the (Computationally-Identifiable) Masses.” *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 1939–48. <https://proceedings.mlr.press/v80/hebert-johnson18a.html>.
- Hopkins, Daniel J., and Gary King. 2010. “A Method of Automated Nonparametric Content Analysis for Social Science.” *American Journal of Political Science* 54 (1): 229–47. <https://doi.org/10.1111/j.1540-5907.2009.00428.x>.
- Jerzak, Connor T., Gary King, and Anton Strezhnev. 2023. “An Improved Method of Automated Nonparametric Content Analysis for Social Science.” *Political Analysis* 31 (1): 42–58. <https://doi.org/10.1017/pan.2021.36>.
- Kadavath, Saurav, Tom Conerly, Amanda Askell, et al. 2022. “Language Models (Mostly) Know What They Know.” *arXiv Preprint arXiv:2207.05221*. <https://arxiv.org/abs/2207.05221>.
- Kim, Michael P., Christoph Kern, Shafi Goldwasser, Frauke Kreuter, and Omer Reingold. 2022. “Universal Adaptability: Target-Independent Inference That Competes with Propensity Scoring.” *Proceedings of the National Academy of Sciences* 119 (4): e2108097119. <https://doi.org/10.1073/pnas.2108097119>.
- King, Gary, and Will Lowe. 2003. “An Automated Information Extraction Tool for International Conflict Data with Performance as Good as Human Coders: A Rare Events Evaluation Design.” *International Organization* 57 (3): 617–42. <https://doi.org/10.1017/S0020818303573064>.
- Knox, Dean, Christopher Lucas, and Wendy K. Tam Cho. 2022. “Testing Causal Theories with Learned Proxies.” *Annual Review of Political Science* 25 (1): 419–41. <https://doi.org/10.1146/annurev-polisci-051120-111443>.
- Mellon, Jonathan, Jack Bailey, Ralph Scott, James Breckwoldt, Marta Miori, and Phillip Schmedeman. 2024. “Do AIs Know What the Most Important Issue Is? Using Language Models to Code Open-Text Social Survey Responses at Scale.” *Research & Politics* 11 (1): 1–7. <https://doi.org/10.1177/20531680241231468>.
- Ornstein, Joseph T., Elise N. Blasingame, and Jake S. Truscott. 2025. “How to Train Your Stochastic Parrot: Large Language Models for Political Texts.” *Political Science Research and Methods* 13 (2): 264–81. <https://doi.org/10.1017/psrm.2024.64>.
- Overos, Henry David, Roman Hlatky, Ojashwi Pathak, et al. 2024. “Coding with the Machines: Machine-Assisted Coding of Rare Event Data.” *PNAS Nexus* 3 (5): pgae165. <https://doi.org/10.1093/pnasnexus/pgae165>.
- Rodriguez, Pedro L., and Arthur Spirling. 2022. “Word Embeddings: What Works, What Doesn’t, and How to Tell the Difference for Applied Research.” *The Journal of Politics* 84 (1): 101–15.

<https://doi.org/10.1086/715162>.

- Rogan, Walter J., and Beth Gladen. 1978. “Estimating Prevalence from the Results of a Screening Test.” *American Journal of Epidemiology* 107 (1): 71–76. <https://doi.org/10.1093/oxfordjournals.aje.a112510>.
- Saerens, Marco, Patrice Latinne, and Christine Decaestecker. 2002. “Adjusting the Outputs of a Classifier to New a Priori Probabilities: A Simple Procedure.” *Neural Computation* 14 (1): 21–41. <https://doi.org/10.1162/089976602753284446>.
- Shimodaira, Hidetoshi. 2000. “Improving Predictive Inference Under Covariate Shift by Weighting the Log-Likelihood Function.” *Journal of Statistical Planning and Inference* 90 (2): 227–44. [https://doi.org/10.1016/S0378-3758\(00\)00115-4](https://doi.org/10.1016/S0378-3758(00)00115-4).
- Sugiyama, Masashi, Shinichi Nakajima, Hisashi Kashima, Paul von Büna, and Motoaki Kawanabe. 2007. “Direct Importance Estimation with Model Selection and Its Application to Covariate Shift Adaptation.” *Advances in Neural Information Processing Systems (NeurIPS)*.
- Tasche, Dirk. 2022. *Class Prior Estimation Under Covariate Shift: No Problem?* <https://arxiv.org/abs/2206.02449>.
- Tasche, Dirk. 2023. *Invariance Assumptions for Class Distribution Estimation*. <https://arxiv.org/abs/2311.17225>.
- Tax, Niek, Lorenzo Perini, Fridolin Linder, et al. 2026. “MCGrad: Multicalibration at Web Scale.” *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2470–81. <https://doi.org/10.1145/3770854.3783954>.
- Tian, Katherine, Eric Mitchell, Allan Zhou, et al. 2023. “Just Ask for Calibration: Strategies for Eliciting Calibrated Confidence Scores from Language Models Fine-Tuned with Human Feedback.” *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. <https://doi.org/10.18653/v1/2023.emnlp-main.330>.
- Wang, Xuezhi, Jason Wei, Dale Schuurmans, et al. 2023. “Self-Consistency Improves Chain of Thought Reasoning in Language Models.” *Proceedings of the 11th International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2203.11171>.
- Weidmann, Nils B., Mats Faulborn, and David Garcia. 2026. “Large Language Models Are Democracy Coders with Attitudes.” *PS: Political Science & Politics* 59 (1): 17–23. <https://doi.org/10.1017/S1049096525101248>.
- Wu, Jiayun, Jiashuo Liu, Peng Cui, and Zhiwei Steven Wu. 2024. “Bridging Multicalibration and Out-of-Distribution Generalization Beyond Covariate Shift.” *Advances in Neural Information Processing Systems 37 (NeurIPS)*, 73036–78. <https://doi.org/10.52202/079017-2325>.
- Wu, Siqi, and Paul Resnick. 2024. “Calibrate-Extrapolate: Rethinking Prevalence Estimation with Black Box Classifiers.” *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)* 18: 1634–47. <https://doi.org/10.1609/icwsml.v18i1.31414>.

- Yang, Yahan, Soham Dan, Dan Roth, and Insup Lee. 2023. “On the Calibration of Multilingual Question Answering LLMs.” *arXiv Preprint arXiv:2311.08669*. <https://arxiv.org/abs/2311.08669>.
- Ziems, Caleb, William Held, Omar Shaikh, Jiaao Chen, Zhehao Zhang, and Diyi Yang. 2024. “Can Large Language Models Transform Computational Social Science?” *Computational Linguistics* 50 (1): 237–91. https://doi.org/10.1162/coli_a_00502.